

James Paul

Senior AI Data Engineer

✉ james.paula615@gmail.com ☎ (304) 403-7273 📍 Lewisburg, West Virginia

PROFILE

Senior AI Data Engineer with 10+ years of experience designing and scaling distributed data platforms and AI-ready lakehouse architectures. Expertise in Apache Spark, Delta Lake, Unity Catalog, MLflow, and streaming frameworks to support large-scale analytics, LLM fine-tuning, and real-time ML workloads across AWS and Azure. Proven track record of optimizing petabyte-scale systems, implementing enterprise-grade governance, high-performance, productionized AI infrastructure.

PROFESSIONAL EXPERIENCE

Databricks, Senior AI Data Engineer

06/2021 – Present

San Francisco, CA - Remote

- Architected enterprise-scale **Lakehouse** platforms during the organization's AI acceleration phase, engineering petabyte-scale pipelines on **Delta Lake and Photon**, supporting GenAI and LLM workloads across global enterprise customers.
- Led production rollout of **Unity Catalog** across AWS and Azure environments, implementing cross-cloud governance with row/column-level security, external locations, workspace isolation, and centralized lineage management.
- Designed fine-tuning and evaluation pipelines aligned with the **Mosaic AI (post-MosaicML integration)** ecosystem, enabling governed LLM deployment using Mosaic AI Model Training and secure model access patterns.
- Built AI agent data backbones aligned with **Agent frameworks and evaluation tooling**, engineering synthetic-data-ready pipelines and automated benchmark datasets to reduce hallucination and improve agent reliability.
- Implemented end-to-end lineage using **Unity Catalog native lineage + external metadata integrations**, providing complete visibility from upstream CRM/OLTP ingestion through BI and ML serving layers.
- Optimized high-concurrency SQL Warehouses and serverless compute using **AQE, Z-Ordering, Delta UniForm, Iceberg interoperability, and Photon acceleration**, reducing enterprise query costs by 38%.
- Partnered with field engineering teams to migrate large-scale Snowflake workloads to **Lakehouse architecture**, modernizing structured + unstructured AI data stacks.

Notion, Senior Data Engineer

03/2018 – 05/2021

San Francisco, CA - Remote

- Scaled behavioral analytics infrastructure during hypergrowth from **1M to 20M+ users**, building ELT frameworks with **Airflow, Snowflake, dbt, and AWS**, enabling near real-time product analytics.
- Engineered data models supporting the company's **B2C2B expansion motion**, tracking workspace virality, template adoption, and free-to-paid conversion metrics.
- Designed collaboration graph pipelines capturing **block-level interaction data** within the all-in-one workspace platform, powering intelligent content surfacing and template recommendations.

- Migrated batch pipelines to streaming-first ingestion using **Kinesis and Spark Structured Streaming**, reducing data freshness SLA from 24 hours to under 10 minutes during pandemic-scale user growth.
- Built experimentation frameworks to measure activation loops and organic acquisition funnels, supporting >90% organic growth dynamics.

Gainsight, Data Engineer

07/2015 – 02/2018

San Mateo, CA - Hybrid

- Built multi-tenant customer health scoring pipelines within a **Salesforce-native customer success platform**, integrating product telemetry, CRM signals, support case metadata, and renewal indicators.
- Engineered secure ingestion frameworks integrating **Salesforce APIs and OAuth-based connectors**, enabling enterprise-grade customer data unification.
- Developed lifecycle and adoption analytics powering **QBR dashboards, churn prediction, and renewal forecasting** for hundreds of SaaS customers.
- Optimized distributed Spark/Hive transformations for high-volume customer success datasets, improving processing efficiency by 27%.
- Implemented compliance-aware segmentation controls to safeguard CRM and subscription metadata across multi-tenant architecture.

SKILLS

Python, Apache Spark, Kafka, MLflow, LLM Fine-Tuning Pipelines, AWS (S3, EC2, Lambda, Kinesis), Azure (ADLS, Synapse), Snowflake, Terraform, Docker

EDUCATION

Master of Science in Computer Science,
University of Texas at Dallas

08/2013 – 05/2015

Bachelor of Science in Computer Science,
University of Texas at Dallas

08/2009 – 05/2013